

PHYSICS

Subject: Physics

Class: Form 1

N° of Periods per week: 2

Annual Teaching Hours: 50

School:

Teacher's Name: Qualification:

TERM	WEEK	MODULE	CHAPTER	LESSON TITLE	Hands-on Activities	Digitalisation of Teaching		№ of Periods			
						Availability of Resources	Resources Used				
First	1	MODULE 1 THE WORLD OF SCIENCE	Introduction to sciences + Scientific contributions and discoveries	First contact with the learners, definition and branches of science.	- Show how straight stick/pencil/ruler will appear when placed inclined in water with one end air and other in water. - Use of a triangular glass prism to produce a spectrum. - Observe writings on a page through a convex lens. - A balloon rubbed so that it sticks to a wall or ceiling when released. (NB: These activities are mainly for observation and curiosity of learners with no explanation at this level.)	Available		1			
				Prominent scientists and discoveries and contributions to improvement in the lives of humans.		Available		1			
	2		Introduction to Physics	Definition of physics and some of its branches. What physicists do and how. Observing nature and seeking comprehension.		Available		1			
			Basic laboratory equipment	Some basic equipment in the Physics laboratory.		Available		1			
	3		Safety laboratory rules	Safety rules for working in the Physics laboratory.		Available		2			
	4		Introduction to measurement	Job opportunities for science students.		Available		1			
				Concepts of simple measurements using measuring instruments.		Available		1			
	5		Physical quantities	Identifying physical and non-physical quantities.		Available		2			
				Some Units of measurement and S.I units.							
	6		Activity of Integration							1	
			EVALUATION							1	

	7	MODULE II MATTER: PROPERTIES AND TRANSFORMATION	Matter	-States of matter and differences between them.	- Measurement of length, using a ruler/tape. - Measurement of diameter of a spherical object by placing it between two parallel sided piece of wood. - Measurement of mass.	Available		1			
	8		States of matter	-interconversion processes.		Available		2			
	9		Measurement of length	- Define length & state its S.I and sub units Measurement of length.		Available		2			
	10		Measurement of mass	- Define mass and state its SI and sub units Measurement of the mass of a body.		Available		2			
	11		Measurement of weight	- Define weight and state its units. - Differentiate between mass & weight. Measurement of weight.		Available		2			
	12		Activity of Integration					1			
	EVALUATION						1				
	Correction and Remediation						1				
	Second		13	MODULE III ENERGY: APPLICATIONS AND USES		Measurement of volume	- Define volume & state its units. -measurement of volumes of liquids, regular & irregular solids.	- Measurement of volume of liquid and solids (regular and irregular shaped). - Measurement of temperature. Application of measurement to determine density of water, piece of stone, piece of cubic or rectangular piece.	Available		1
			14			Measurement of Density	- Define density as the mass per unit volume and state its units.		Available		1
			15			Measurement of Temperature	-Define temperature and state its S.I and sub units/conversion.		Available		1
						Using information on products	-safety rules on products/ materials.		Available		1
-using information.											
16		Forms of energy	-Definition, forms, sources of energy.	- Use of candle wax marks placed on metal and non-metal and the other end held close to a candle flame to demonstrate good and bad conductors. - Convection: Water is heated in a beaker, plastic transparent cup or polythene bag.	Available		1				
17		Energy needs	-Daily applications of energy.		Available		2				
			-common devices that use different forms of energy and principle of energy conservation.		Available		2				
		Solar energy	- Components and uses of solar energy.								
Chemical energy		-sources and uses of chemical energy.	Available								
18		Activity of Integration					1				
EVALUATION						1					
Correction and Remediation						1					
19		Electrical energy	-sources and uses of electrical energy.			Not yet		1			
20		Thermal energy	-sources and uses of heat.			Not yet		1			
			-Heat transfer by conduction (conductors / insulators).	Not yet			1				

				-Heat transfer by convection / application. (ventilation) -Heat transfer by radiation (good and bad absorbers/emitters).	burning candle flame. - Soaking a piece of paper and holding it close to a burning candle. - Light a bulb using a cell or battery.				
	21		Forces and motion	Definition and effects of forces.		Not yet		1	
				-definition/types of motion (straight line, circular and to and fro).		Not yet		1	
	22			-safety rules: uses of seat belts, low speed/road signs.		Not yet		2	
	23	MODULE IV HEALTH EDUCATION	Sound	Definition and production.	- Measurement body temperature using a clinical thermometer. - Making of any local instrument (eg tomato tins plus metal wires) to produce sound such that loudness can be varied.	Not yet		1	
				The ear and sound perception.		Not yet		1	
	24		Activity of Integration					1	
			EVALUATION					1	
	25		Correction and Remediation					1	
			Detection of sound	-Effects of loud sound and prevention		Not yet		1	
	26		Temperature	-measurement of temperature of the human body (clinical thermometer). -normal and abnormal body temperatures.		Not yet		1	
			Sports and physics	Body posture: importance of good posture to the body.			Not yet		1
Third	27	MODULE V ENVIRONMENTAL EDUCATION AND SUSTAINABLE DEVELOPMENT	- RADIATIONS	-Harmful waste & background radiation. -Handling radioactive substances.	- Two boxes are designed from glass or transparent plastics with one completely sealed leaving a possibility to insert a thermometer while the other has an opening. The two boxes are exposed to the sun for about 30 minutes, the readings of the thermometers can be used to explain the greenhouse effect	Not yet		2	
	28		-global warming And climate change	-greenhouse effect (concept/causes). -climate change. -environmental sustainability.		Not yet		2	
	29	MODULE VI TECHNOLOGY	Common tools	Machines (identification and uses).	- Drawing the 6 faces of an object such as a digital multimeter separately.	Not yet		1	
			Maintenance	-lubrication, cleaning and repairs using the tools.		Not yet		1	
	30		Activity of Integration					1	
			EVALUATION					1	
	31		Correction and Remediation					1	
			Technical drawing	Instruments used/ sample drawing.		Not yet		1	
	32 - 36	GENERAL REVISION / END OF YEAR EXAMINATION							10

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Subject: Physics

Class: Form 2

N° of Periods per week: 2

Annual Teaching Hours: 50

School: Teacher's Name: Qualification:

Term	Week	Module	Chapters	Lesson title	Hands-on Activities	Digitalisation of Teaching		№ of Periods	
						Availability of Resources	Resources Used		
FIRST	1	Module 1: The world of science	First contact with students and presentation of scheme of work.		- Loading a helical spring to obtain a table of mass and corresponding extension and calculate the mass per unit length. - Provide rectangular wooden blocks of same material, for learners to measure length, breath, height and mass and use to calculate density.			2	
			1: Scientific method part 2	1.0 Collecting data-importance of data.					1
	1.2 Interpreting data and concluding.						1		
	1.3 Predicting and evaluating.							2	
	2: Simple application of measurements		1.4 Planning (recall basic quantities units and instruments).				2		
			1.5 Measurements of speed and units.						2
		1.6 Measurement of density and Units.			2				
	5	Module 2: matter, Properties and transformation	3: Change of state	2.1 Physical states of matter. 2.2 Characteristics of matter in the different states.		- Measuring the temperature of a room, water and human body. - Allowing ice to melt while in contact with another object whose temperature we can measure. - Allowing spirit to evaporate while in contact with a body whose temperature we can measure.	Available		2
	6		Activity of Integration						1
			EVALUATION					1	
			Correction and Remediation					1	
	7		4: Temperature	2.3 Temperature – measurement and units and effects on matter.	Available			1	
	8		5: Thermal and electrical insulation	2.4 Thermal and electrical insulation.	Available			2	
	9		6: Action of heat and electrical energy	2.5 Action of heat on materials. 2.6. Action of electrical energy on materials.	Available			2	
	10		Module 3: energy, value and uses	9: Energy needs of human beings	3 Sources and uses of energy. transmission of energy.		- Light a candle so that it burns and we can show that it gives out heat which can be used for heating, drying, cooking as light for seeing.	Not yet	
	11	10: Renewable energy		-Solar panel for heating. -Other sources.	Not yet			2	
	12	Activity of Integration				1			
		EVALUATION				1			

SECOND	13		Correction and Remediation		- Use a photovoltaic plate to light a small bulb to identify forms of energy. - Power a small radio with batteries for it to emit sound.	Not yet		1	
			11: Electricity	Electricity for the home and the simple electric circuit.				1	
			14	12: Light				Sources of Light.	1
				12: Light				- Types of light receivers.	1
								-Beams and shadows.	1
	15	13: Energy exchange	Linking one form of energy to other forms.	Not yet		2			
	16	14: Motion	Distance, time and speed.		Not yet		2		
	17	Module 4: Health education	15: Distribution of pressure in a liquid	4. Average blood pressure.	- Use a syringe and plastic bottles with small holes to demonstrate liquid pressure when pushed using a piston or driven by a height difference.	Not yet		2	
	Activity of Integration			1					
	EVALUATION			1					
	Correction and Remediation			1					
	18		16: Muscle stress	Sports and physical education.				Not yet	1
	19		17: Types of lenses and their uses.	The eye as an imaging device and use of lenses to aid eyes with vision defects.				Not yet	2
	20	Module 5: Environmental Protection and sustainable development	18: Radiation	5.0 Introduction 5.1 Radiation emitted into the atmosphere. -Cosmic waves from the sun.	Using transparent glass or plastic to design a box in which there is a possibility to insert a thermometer and measure temperature with time after exposing to solar radiation.	Not yet		2	
	21		19: Weather and communication	5.1 The Greenhouse Effect.				Not yet	2
	22			5.3 Global warming.				Not yet	2
	23			Activity of Integration					1
	24			EVALUATION					1
				Correction and Remediation					1
	5.4 Climate change.			Not yet				2	
THIRD	25	Module 6: Technology	22: Introduction to technology	6.0: Introduction to technology Review of form one.	- Use a screw driver to drive a screw - Use a tester to identify a live terminal of a source. - Draw a 3-D diagram of a multimeter.	Not yet		2	
	26		23: Project	6.1 Fabrication of common instruments.				Not yet	2
	27		24 Care and maintenance	6.2 Care and maintenance. Principle of functionality of some common appliances.				Not yet	2
	28		25: Technical drawing	6.3 Technical drawing.				Not yet	2
	29								
	30 - 36		GENERAL REVISION / END OF YEAR EXAMINATION						14

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Subject: Physics

Class: Form 3

N° of Periods per week: 2

Annual Teaching Hours: 50

School: Teacher's Name:

Qualification:

TERM	WEEK	MODULE	CHAPTER	LESSON TITLE	Hands-on Activities	Digitalisation of Teaching		№ of periods	
						Availability of Resources	Resources Used		
FIRST	1	MODULE 1: INTRODUCTION TO MECHANICS	1. Physical quantities	First contact with the students					02
				- Definitions, examples and units of physical quantities. - Scalar physical quantities and vector quantities.	- Measure the length, width, and height of a box using a metre rule and use these measurements to determined its volume. - Measure the diameter of a spherical object by placing it between two parallel sided piece of wood and use the measurement to determine its volume.	Available			
				- Prefixes and use of standard form. - Name some basic equipment used in the study of force: newton meter, balance, springs, masses etc.		Available			
				- Safety rules for working with different equipment. - comparison between mass and weight.		Available		02	
				Measurement of mass/Conversions between different units of mass.		Available			
				2		Measurement of volume, temperature and time.	Available		02
	3	MODULE II: MATTER: PROPERTIES AND TRANSFORMATION	2. Density	Definition / calculation / unit of density.	- Measure the density of a liquid, regular and irregular solids use the values obtained to determine whether the liquid or solid will float on water. - Use a collapsing can to demonstrate atmospheric pressure. - Use a syringe and plastic bottles with small holes to demonstrate the characteristics of pressure in liquids.	Available		02	
	4			- Describe and carry out experiments to measure the density of regular & irregular objects. - Applications of the density of a material in engineering works.		Available			
				3. Pressure		- Definition / calculation / unit of pressure. - Factors that affect pressure in solids. - Application of pressure in solids.	Available		
			Activity of Integration					01	
	6		EVALUATION					01	
	7		Correction and Remediation					Available	
			- Pressure in liquids. - Calculation / Factors affecting of pressure in liquids. - Characteristics of pressure in liquids.						

				– Applications; Hydraulic machines.	- Load a helical spring and a rubber band separately with standard masses to obtain tables of mass and corresponding extension to: - Demonstrate Hooke’s law. - Distinguish between materials that obey Hooke’s law and those that do not.			02	
	8			– Atmospheric pressure and weather. – Demonstration/ Measurement/Application of atmospheric pressure. – Pressure and health (Effects of high/low pressure).		Available			
				- Effects of pressure on boiling point. -Application of high pressure.		Available			
	9			4. Elasticity and Hooke’s law		– Definition of elasticity. – Elastic and non-elastic materials. – Hooke’s Law and elastic limit.	Available		
	– Sketch F – e graphs within the elastic limit. – Experimental demonstration of Hooke’s law. – Describe situations in which Hooke’s law applies.	Available							
	10	MODULE III: ENERGY: APPLICATIONS AND USES	5. Forms of energy	– Definition / Forms of energy.	- Use a photovoltaic plate to light a small bulb to identify forms of energy. - Light a candle so that it burns and a small radio powered with batteries to demonstrate conversion of energy. - Use a burning candle and a small bulb powered by a photovoltaic plate to distinguish between renewable and non-renewable energy sources. - Use a spring or a rubber band to project a small object to demonstrate conversion of potential energy to kinetic energy. - Mount systems of one, two, three, four or five pulleys and use to lift loads.	Available		02	
			6. Sources of energy	– Sources of energy. – Renewable and non-renewable energy sources.		Available			
			7. Energy Transformations	– The law of conservation of energy / application. – Transducers and energy flow diagrams.		Available			
	11			– Energy transformations. – Calculations of potential (gravitational and elastic) and kinetic energies.		Available		02	
			8. Work	– Definition / Calculation of work involving force and displacement (in the same direction only). – Examples of situations where work is done.		Available			
	12		Activity of Integration					02	
			EVALUATION						
			Correction and Remediation					02	
	13		9. Power	– Definition / calculation of power. – Power ratings of some devices e.g. electric iron, light bulbs, electric motors etc.		Available			
				– Definition / advantages of a machine.		Available			

SE CO ND			10. Simple Machines	- MA / VR / Efficiency. - Relationship between the three.	- Use a simple torch and slits to produce rays and beams. - Use a simple torch, slit(s) and mirrors to demonstrate reflection of light. - Show how straight stick/pencil/ruler will appear when placed inclined in water with one end air and other in water. - Use a simple torch, slit and glass block to demonstrate refraction of light, critical angle and total internal reflection. - Use a burning candle and a convex lens to demonstrate images formed by converging lenses. - Use a prism to demonstrate dispersion of white light.				
	14			- Simple machines (Lever, inclined plane and pulley, hydraulic machines) / Experiments and calculations .		Available		02	
	15		Optics 10. Reflectio n of light	- Description / Propagation of light. - Rays and types of beams. - Luminous and non-luminous sources of light. - Transparent, translucent and Opaque objects. - Shadows and applications.		Not yet		02	
				- Definition / laws of reflection of light. - Experiment to show that i = r		Not yet			
				CHRISTMAS BREAK			CHRISTMAS BREAK		
	16			- Image formation by plane mirrors. - Characteristics of images formed by plane mirrors. - Uses / Simple calculations on plane mirrors. - Curved mirrors (Types and applications).		Not yet		02	
	17		11. Refractio n of light	- Identification of refraction phenomena in everyday life / Definition of refraction.		Not yet			
				- Laws of refraction. - Refractive index and speed of light.		Not yet		02	
				- Experiment to determine refractive index of a glass block. - Effects of refraction of light in everyday life. - Real/apparent depth and refractive index.		Not yet			
				Activity of Integration					
	18		EVALUATION					02	
	19		Correction and Remediation						
12. LENSES		- Total internal reflection (TIR). - Conditions / Daily occurrence / Definition of total internal reflection. - Ray diagram to show TIR.	Not yet		02				
		- Use of TIR / Solve problems using $sinc = \frac{1}{n}$	Not yet						
20		- Lens action/Types of lenses/Definition of terms:	Not yet		02				

	21			– Ray diagrams to illustrate the meaning of principal foci, for converging & diverging lenses. – Measurement of focal length of a converging lens.		Not yet		02	
	22			– Image formation by converging/diverging lenses. – The lens formula and magnification, $m = \frac{v}{u}$		Not yet		02	
	23			13. Dispersion of light.		– Definition / Demonstration of dispersion.	Not yet		02
						– Formation of a pure spectrum from white light. – Natural occurrence of the dispersion.	Not yet		
	24						Activity of Integration		
			EVALUATION						
THIRD	25	MODULE IV: Projects and Elementary Engineering	14. Understanding how to go about a Project.	– Technical drawing: General notions involved. – 2-dimensional diagrams/orthogonal representation.	- Draw a two-dimensional diagram of a wall of the classroom containing the door and windows.	Not yet		02	
				– Technical Project Types /Elements involved. – Definition/Steps involved in planning a project.		Not yet			02
	26			– Study the advantages offered by the project; – Feasibility Studies. – Theoretical knowledge of a simple technical object Different energy components when using the instrument;		Not yet		02	
				27		– Technical diagram / studies. – Identification of the movement of certain parts with respect to others (guide, reducing friction).	Not yet		02
	28		15. Investigating Forces	– Definition of new words. – Realisation of a project. -Understand the uses of measuring instruments and their limitations. Understand why and how things move. – Understand the relationship between mass and volume through measurements.		Not yet		02	
				EASTER BREAK		EASTER BREAK			
	29			– Observing colours of light on a CD or DVD and suggesting reasons for their appearance.		Not yet		02	

				– Observe a pen or pencil in a glass of water and explain why there is an image seen with the object.				
	30			– Use the image to estimate the refractive index of water.		Not yet		02
	31-35	GENERAL REVISION / END OF YEAR EXAMINATION						

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School: Teacher's Name: Qualification:

Term	Week	Module	Chapters	Lesson titles	Hands-on Activities	Digitalisation of Teaching		No Of periods		
						Availability of Resources	Resources Used			
FIRST	1	MODULE 1 ENERGY: APPLICATION AND USES	First contact with students and presentation of scheme of work						3	
			Heat	1.1 Concept of heat and temperature. 1.2 Measurement of temperature.	- Use a thermometer to measure temperature. - Determine the specific heat capacity of water using an electric heater. - Use of candle wax marks placed on metal and non-metal and the other end held close to a candle flame to demonstrate good and bad conductors. - Water is Heat water in a beaker, plastic transparent cup or polythene bag to demonstrate convection. - Candle wax on a metal or a soaked piece of paper is held at a distances from a burning candle flame to demonstrate radiation.	Available				
	Thermometry		1.3 Thermometry. 1.4 liquid-in-glass thermometer. 1.5 Calibration of temperature scales using fixed points.	Available			3			
				1.6 Cinical and normal laboratory thermometers.		Available				
			Calorimetry	1.1 Calorimetry. 1.2 Heat capacity and specific heat capacity, c. 1.3 Measurement of specific heat capacity, c for solids and liquids, Calculation involving $Q = mc\Delta\theta$		Available			3	
	Latent Heat		1.4 Latent Heat and specific Latent heat. 1.5 Cooling effect.	Available			3			
	Heat transfer		1.6 Heat Transfer. -Conduction, Convection and Radiation.	Available						
	Expansion		1.7 Thermal Expansion; -The bimetallic strip and its principle.s -Radiant energy converters.	Available			3			
	6		Activity of Integration						3	
			EVALUATION							
		Correction and Remediation								
	7	MODULE 2: WAVES	Properties of Waves	2.1 Definition and classification of waves. 2.2 Properties of waves: Reflection, Refraction, Diffraction and Interference. 2.3 Calculations using the equation $v=f\lambda$.	- Use a slinky coil to produce transverse and longitudinal waves. - Use a ripple tank or water in a basin to produce water waves and demonstrate properties of water waves.	Available		3		
	8		Stationary Waves	2.4 Stationary waves. 2.5 Harmonics and Overtones. 2.6 Relationship between inter-node distance and		Not yet		3		

				wavelength.					
	9		Sound Waves	2.7 Production and transmission of sound. 2.8 Characteristics of sound (Amplitude, frequency, and pitch, quality and overtones. Frequency limit of audibility).		Not yet		3	
	10			2.9 Measurement of the speed of sound (echo method and gun firing method). Application of echo in sea bed. 2.10 Calculation of speed of sound in air.	Not yet		3		
	11		Vibration in strings	2.11 Vibrating strings, Relationship between frequency and length, frequency and mass per unit length, frequency and tension.	Not yet		3		
	12		Activity of Integration						3
EVALUATION									
Correction and Remediation									
SECO ND	13		Forced vibration and Resonance	Forced vibration on a string and in a tube Define and explain resonance. -State importance and applications of resonance including measurement of speed of sound.		Not yet		3	
	14		EM Spectrum	- Relative positions of radiation on EM spectrum, in terms of wavelength and frequency. - Methods of production and detection. - Properties and used. - Health hazards caused by EM waves.		Not yet		3	
	15		MODULE 3 ELECTRICAL ENERGY	ELECTROSTATICS	3.1 Charges and their origins. 3.2 Types of charges and the Basic law of electrostatics. 3.3 Coulomb’s law 3.4 Testing for charge with electroscope. 3.5 Charging and discharging. 3.6 Separation of charges.	- A balloon rubbed so that it sticks to a wall or ceiling when released to demonstrate charging by friction.	Not yet		3
	16			The coulomb	3.7 conductors and insulators in relation to spreading of charge on charged objects.		Not yet		3
	17				3.8 Force between charges, Positive and negative charges,		Not yet		3

				Application of electrostatics industrially eg photocopying, painting.					
18		Activity of Integration							3
		EVALUATION							
		Correction and Remediation							
19		Current Electricity	3.9 Introduction 3.10 Electricity and charge movement. Use of meters in electric circuits, how they are connected and reasons.		Not yet			3	
<u>20</u>		Emf	3.11 Emf and pd sources of emf.		Not yet			3	
21		Electric circuits	3.12 Energy consumption. W=QV and P=VI	- Building simple electrical circuits with small torch bulbs of know resistances: Connecting at least two bulbs in series or parallel and calculate their combine resistance.	Not yet			3	
22			3.13 ELECTRIC CIRCUITS. -components of a circuit network.		Not yet			3	
23			3.14 Ohms’ law and Resistance. 3.15 Circuit network. (No mention of Kirchhoff’s rules) -Parallel and series connections. Calculations in circuits.		Not yet			3	
24			Activity of Integration						3
			EVALUATION						
			Correction and Remediation						

Term	Week	Module	Chapters	Lesson title		Digitalisation of Teaching		No Of periods
						Availability of Resources	Resources Used	
THIRD	25	MODULE III ELECTRICAL ENERGY	Power in ac and dc	3.16 DC and AC, Definition Transportation of electric current. 3.17 Calculation of power dissipated.		Not yet		3
	26			3.18 Calculation of power consumed at home (KWH).		Not yet		3
	27		House wiring	3.19 House wiring, Ring circuit and linear circuit in house wiring.		Not yet		3
	28			3.20 Fuse and selection, Safety precaution.		Not yet		3
	29		CRO	3.21 The cathode ray oscilloscope.				3

	30		Activity of Integration						3
			EVALUATION						
			Correction and Remediation						
	31	MODULE IV PROJECTS AND ELEMENTAY ENGINEERI NG	Technical Drawing	Realization of a technical project. 1 Technical drawing. 2 Reading of technical drawing.	Draw a three dimensional diagram of a wall of the classroom containing the door and windows.	Not yet		3	
	Section view Building plans		1.2. Cross section of an Object.	Not yet			3		
			List the various ways of viewing an object. e.g top – view.						
Reading of the plan of a construction sheet. Drawing and giving of dimensions of a building.									
33				Not yet		3			
34 - 36	GENERAL REVISION / END OF YEAR EXAMINATION								9

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Subject: Physics

Class: Form 5

N° of Periods per week: 3

Annual Teaching Hours: 63

School: Teacher's Name: Qualification:

Term	Week	Module	Chapters	Lesson title	Hands-on Activities	Digitalisation of Teaching		No of Periods
						Availability of Resources	Resources Used	
FIRST	1	Module 1: FIELDS: MAGNETIC FIELDS AND THEIR EFFECTS	First contact with students and presentation of scheme of work					3
	Magnetic Field		1.0: Magnets 1.1 Introduction to magnetism- magnetic and non-magnetic substances. -Identification of magnets.	- Use a magnetic compass to identify non-magnetic, magnets and ferromagnetic materials. - Use iron filings to trace the magnetic field pattern round a magnet. - Use a thread to hang a bar and use it to identify use to identify the earth's magnetic North. - Produce a solenoid using a nail and copper wire connected to battery and use it to pick up small magnetic materials. - Move a bar magnet at different rates towards a coil connected to a bulb.	Available			
			1.2 Law of magnetism - applications of magnets. - Making magnets and care for magnets. -Hard and soft magnetic materials.		Available		3	
			1.3 Magnetic Field. 1.4 Magnetic flux pattern; -defining magnetic flux -Drawing of magnetic field lines for; bar magnet, the earth and the horse-shoe magnet. -flux pattern around two poles of magnets near each other.		Available		3	
			Magnetic effect of current 1.5 Magnetic effect of current -Magnetic field pattern of a straight conductor carrying current. -direction of field lines: current carrying solenoid and factors which affect field strength.		Available		3	
5		- Force on a current carrying conductor placed in a magnetic field. - Factors which determine the size of the force. Principle of electric motor.	Available		3			

	6		Activity of Integration						3
	EVALUATION								
	Correction and Remediation								
	Electromagnetic Induction and Alternating Current		1.6 Electromagnetic Induction - Introduction to electromagnetic induction. - Faraday's law.		Available		3		
			- Lenz laws. - Conservation of energy.		Available		3		
			1.7 Alternating Current. Mutual inductance.		Available		3		
		- The transformer, - Energy losses and remedies - Turn ratio		Available		3			
	11		- Efficiency - Practical transformers and power transmission - Applications of transformers.		Available		3		
	12	Activity of Integration						3	
		EVALUATION							
		Correction and Remediation							
SECOND	13	MODULE 2 ENVIRONMENTAL PROTECTION: Modern Physics and basic electronics.	The Atom	2.0 The nuclear model of the atom; - the composition of the atom. The electron (Q=Ne)		Available		3	
	14			2.1 The nuclear model of the atom - the composition of the atom The electron (Q=Ne)		Available		3	
	15		The Nucleus	2.2 The Nucleus. Nuclear stability. 2.3 Radioactivity and decay equations.		Available		3	
	CHRISTMAS BREAK								
	16		Radioactivity	2.4 Radioactive decay α , β and γ . And their properties including behaviour in electric field, magnetic field and cloud chamber.		Available		3	
	17			2.5 The concept of half-life -The importance and use of isotopes. - Background radiation.		Available		3	

	18		Activity of Integration					3	
	EVALUATION								
	Correction and Remediation								
	19		2.6 uses of radioactive isotopes: eg in medicine, agriculture. 2. 7: safety and hazard of radioactivity.		Available		3		
	20	Basic electronic s	2.8 Semiconductors; -intrinsic and extrinsic -P-type and N-type P-n junctions and rectification.		Available		3		
	21	Module 3: MECHANICS	Physical quantities	3.1 Vector and scalar physical quantities 3.2 Forces; -Type of forces. -Resolution of forces. -Free body diagrams. -Archimedes Principle.	Use magnets to demonstrate non-contact forces. Pulling very fast an A4 paper on which an object has been placed to demonstrate Newton’s first law.	Not yet		3	
	22			3.3 Mass and weight. 3.4 Turning effect of forces. -Moments, couples and applications.		Not yet		3	
	23		Motion	3.5 Linear motion; - Distance, displacement, Speed, velocity, acceleration. -motion graphs.		Not yet		3	
	24			Activity of Integration					3
				EVALUATION					
			Correction and Remediation						
	THIRD		25		3.6 Uniform motion; -Equations of uniformly accelerated linear motion and simple calculations. - Free fall and gravity. -Experiment to determine g, by free fall.		Not yet		3
			26	Newton’s law	3.7 Linear momentum -Interaction and conservation of momentum. -Principle of conservation of momentum and real life situations.		Not yet		3

	27			3.8 Newton's laws of motion		Not yet		3	
	28	Module 4: PROJECTS AND ELEMENTARY ENGINEERING	Preservat ion of appliances	Preservation and Maintenance of appliances.	- Dismantle an electric motor and re- assemble. - Use the electric motor to pull a load.	Not yet		3	
	29			The essential elements in a repair box. (give the name, role and method of application).					
	30		Maintena nce of appliances	Understand the labelling on appliances.		Not yet		3	
	31			Techniques of dismantling and assembling of appliances (FOLI AND LIFO) techniques.					
	32 - 36		GENERAL REVISION/ END OF YEAR EXAMINATION						15

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Subject: Physics

Class: Lower Sixth Science

N° of Periods per week: 8

Annual Teaching Hours: 270

School: Teacher's Name: Qualification:

Term	Week	Module	Chapter	Lesson title	Digitalisation		Duration	Module	Chapter	Lesson title	Digitalisation		Duration	
					Availability	Used					Availability	Used		
I	1	Module 1: Physical quantities	First contact with students and presentation of scheme of work/Advanced Level course outline.				02H	Module 3: Energetics	First contact with students and presentation of scheme of work/Advanced Level course outline				02H	
			1.1 Physical quantities	Physical quantities, base and derived physical quantities/units, SI units.	YES		02H		3.1 Temperature		Thermal equilibrium, zeroth law and thermometric properties.	YES		02H
	Dimensions, physical equations and homogeneity of physical equations.			YES		04H								
	Experimental physics. Experimental techniques, approach, accuracy, sensitivity error and precautions.													
	3		2.1 Kinematics	Motion, distance/displacement, speed/velocity, Linear momentum, acceleration.		YES			04H	3.2 Energy Transfer		Heat capacity and specific heat capacity.	YES	
				Graphs of motion in one dimension and equations of uniformly accelerated motion.										
	4			Motion under gravity and experiment to measure acceleration due to gravity.		YES		04H	Experiments to measure specific heat capacity.		YES		02H	
				Projectile motion **This could still be taught under motion in the gravitational field.		YES		04 H						
	5			2.2 Dynamics and force	Meaning and nature of forces, types and classification. Centre of gravity, centre of mass, free-body diagrams and resultant of coplanar forces.		YES		04H	Latent heat and specific latent heat.		YES		02H

				Turning effect of forces, moment and couples, integration exercise.						Experiments to measure specific latent heat of fusion and specific latent heat of vaporization.	YES		
	6			REVISION AND EVALUATION			04H			REVISION AND EVALUATION			02H
	7			Equilibrium of objects acted upon by a number of coplanar forces.	YES		04H			Heating and cooling curves, integration exercises	YES		02H
	8			Newton's first and second Laws, impulse, experimental investigations of $a \propto F$ and $a \propto \frac{1}{m}$.	YES		04H		3.2 Energy Transfer	Thermal conduction and mechanism, temperature gradient.	YES		02H
	9	2.3 Newton's Laws of motion and momentum	Newton's Third law of motion, the law of conservation of linear momentum from Newton's second and third laws.	YES		04H		Thermal conductivity,		YES		02H	
	10		Law of conservation and experiment to investigate the law of conservation of linear momentum.	YES		04H		Comparing electrical conduction to thermal conduction, experiment to compare thermal conductivity of different materials.		NO		02H	
	11		Elastic and inelastic collisions.	YES		04H		Thermal convection and radiation, mechanisms		YES		02H	
			Explosions, head-on and oblique collisions.					3.3 Electrical energy					Electric current, electric potential difference, drift velocity.
	12							INTEGRATION EXERCISES AND EVALUATION					
II	13	2.4 Work, energy and power	Work, potential energy and kinetic, Law of conservation of mechanical energy.	YES		04H		3.3 Electrical energy	Electrical resistance, resistor colour codes, Ohm's Law and resistivity.	YES		02H	
	14		Work-kinetic energy theorem, efficiency, power.	YES		04H			Experiment to determine resistivity, temperature coefficient of resistance.	YES		02H	
	15	2.5 Circular motion	Period and frequency, angular speed and velocity, centripetal	YES		04H			Resistor networks and the potential divider.	NO		02H	

				acceleration and centripetal force, consolidation examples.										
			CHRISTMAS BREAK						CHRISTMAS BREAK					
16			Motion in a vertical circle, the conical pendulum, cornering and banking.	NO		04H			Ideal and non-ideal ammeters and voltmeters in circuit.	NO		02H		
17	2.6 Simple harmonic motion		Meaning and equations.	YES		04H	3.3 Electrical energy		Electromotive force, terminal p.d. and internal resistance.	NO		02H		
18					HARMONISED EVALUATION									
19	2.6 Simple harmonic motion		Energy changes in simple harmonic motion, examples of simple harmonic oscillators.	NO		04H		3.3 Electrical energy	Kirchhoff's Laws.	NO		02H		
20			Mechanical oscillations and resonance.			04H			Potentiometer.	NO		02H		
21	Module 4: Matter, effects of energy and application	4.1 Solids and liquids	Differences in the molecular properties of solids, liquids and gases, molecular spacing, intermolecular force vs separation curves, potential energy vs separation curves.	NO		04H	Option 1: Energy Resources and environmental Physics		Wheatstone bridge circuit, consolidation exercises.	NO		02H		
22			Elasticity and Young modulus energy stored in a stretched wire.	NO		04H		3.4 Energy resources and environmental Physics	Classification of energy sources, functional energy, efficiency of conversion.	NO		02H		
23			Experiment to determine Young Modulus, consolidation exercises.	NO		04H			Hydroelectricity and wind energy.	NO		02H		
			24	REVISION AND EVALUATION				04H	REVISION AND EVALUATION			02H		
III	25	Surface tension and capillarity, Angle of contact, experiment to measure surface tension of water.	NO		04H		Solar energy and tidal energy.	NO		02H				
	26	4.2 Gases and thermodynamics	Brownian motion in gases, gas laws and the ideal gas equation.	NO		04H		Biomass, geothermal energy and wave energy.	NO		02H			

	27	4.2 Gases and thermodynamics	Kinetic theory of ideal gases, derivation of $P = \frac{1}{3} \rho \overline{c^2}$ and $k.e. = \frac{1}{2} m \overline{c^2} = \frac{3}{2} kT$. Distribution of molecular speeds, Real gases and Andrew's experiment.	NO		04H		3.4 Energy resources and environmental Physics	Fossil fuels and nuclear fuel.	NO		02H
	28		First law of thermodynamics; isochoric, isobaric, isothermal and adiabatic processes.	NO		04H			Simple structure of the atmosphere, ozonosphere and the ionosphere, destruction of the ozone layer.	NO		02H
	29		Second law of thermodynamics, basic function of heat engines, entropy, Consolidation examples.	NO		04H			Greenhouse effect, global warming and climate change.	NO		02H
			EASTER BREAK									
	30 - 36	GENERAL REVISION AND END OF YEAR EVALUATION										

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Subject: Physics

Class: Upper Sixth Science

N° of Periods per week: 8

Annual Teaching Hours: 270

School: Teacher's Name: Qualification:

Term	Week	Module	Chapter	Lesson title	Digitalisation		Duration	Module	Chapter	Lesson title	Digitalisation		Duration
					Availability	Used					Availability	Used	
I	1	Module 5: Field Phenomena		First contact with students and presentation of scheme of work/USS course outline • Newton's law of universal gravitation and inverse square relation.	YES		02H	Module 6: Waves around us.		First contact with students and presentation of scheme of work/USS course outline. 6.1 6.1 MECHANICAL WAVES: Classification of Waves: mode of propagation and medium of propagation with specific examples.			02H
			5.1 GRAVITATIONAL FIELDS	• Kepler's laws, Qualitative description of the Earth's gravitational field. • Field Strength.	YES		04H						
	2			• Variation of g inside and outside the Earth. • Gravitational potential and gravitational potential energy.	YES		04H		6.1 MECHANICAL WAVES	• The progressive wave and equation. • Graphical representation of waves.	YES		02H
	3			• Escape velocity Orbital speed and geostationary satellites. Orbital speed and movement of satellites with geostationary satellites as example.	YES		04H			Properties of Waves Production of waves, reflection, refraction, diffraction: Interference. Single slit pattern	YES		02H
	4		ELECTRO STATICS + ELECTRIC FIELDS	. Motion in the gravitational field: Projectile motion. **Check if this was not treated under mechanics in LSS.	YES		04H			Double slits and multiple slits interference patterns and measurement of wavelength of a wave.	NO		02H
	5			• Electric charge and current • Good and bad conductors of electricity. • Charging by friction, contact, induction and by chemical action.. • Point action and the lightening conductor.	YES		02H + 02H			• Properties of Waves: Polarization-meaning and production of plane polarized waves.	NO		02H
	6			REVISION AND EVALUATION			04		REVISION AND EVALUATION				02H

	7			• Coulomb's law and inverse square nature. Dependence of size of force on medium. • Electric field and field strength.	YES		04H	3.2	• The factors affecting the speed of transverse waves on taut strings and wires	NO		02 H
	8			• Electric potential and work done in moving a charge in an electric field. • Electric field dipole and torque.	YES		04 H		Doppler Effect for Sound in air with specific cases of moving source/ stationary observer and moving observer with stationary source.	YES		04 H
	9		CAPACITORS	• Identification of capacitors and circuit symbols. • Measurement of capacitance. • Factors which affect capacitance. • Relative permittivity.	YES		04H		Stationary waves and characteristics.	YES		02H
	10			• Combination of capacitors (series & parallel) • Energy/charge stored in a different combinations of capacitors connected to a dc power supply	YES		04H		Measurement of the speed of sound in air.	NO		02H
	11			• Charging and discharging capacitors through resistors and time constant. The equations for charging/ discharging and interpretations at $t=0$ and as $t \rightarrow \infty$.	YES		04H		• Electromagnetic waves and their characteristics. • EM-spectrum; production, detection and uses of different sections	YES		02H
	12			REVISION AND EVALUATION			04H	REVISION AND EVALUATION			02	
II	13		5.4 MAGNETIC FIELDS	• The magnetic field. • Magnetic flux density and its units (the tesla).	YES		04H	3.2	• X-rays, production and uses. • Meaning and application of	NO		02H

			- Field patterns of current-carrying conductors (straight wire, plane circular coil, solenoid) - Force on a current-carrying conductor in a uniform magnetic field. - The direction of the force from Fleming's left hand rule.						plane polarized em waves.			
14		5.4 MAGNETIC FIELDS	- Forces on objects in cross-fields - Torque on a rectangular coil within a uniform magnetic field ($T = NAI\mathbf{B}$) - The principle of the electric motor (dc and ac types)	NO		04H			- Light sources. - Optical transmission grating with normal incidence.	NO		02H
15			- Biot-Savart law - Ampere's law - Magnetic flux density within a long solenoid, long straight wire, and plane circular coil ($\mathbf{B} = \mu_0 n \mathbf{I}$, $\mathbf{B} = \mu_0 \mathbf{I} / 2r$, $\mathbf{B} = \mu_0 \mathbf{I} / 2\pi r$). Force between two current-carrying conductors.	NO		04H			- Multiple slit diffraction. - Reflection and refraction at plane interfaces.	NO		02H
			CHRISTMAS BREAK						CHRISTMAS BREAK			
16			- Force on a moving charge in uniform magnetic field - Measurement of specific charge (e/m_0) - The Hall effect - dia, para- and ferro-magnetic materials - Magnetic Shielding - The Lorentz force	NO		04H			- Laws of refraction. - Refractive index.	YES		02H
17		ELECTRO MAGNETIC INDUCTION	- Faraday's and Lenz's laws of electromagnetic induction. - Induced e.m.f. across a conductor moving with	NO		04H		3	- Dispersion. - Total internal reflection and critical angle.	YES		02H

			velocity v, through uniform magnetic field. • Simple DC and AC generator principles.						• Lenses: principal focus, focal length.			
	18		REVISION AND EVALUATION			04H	REVISION AND EVALUATION			04H		
	19	ELECTRO MAGNETIC INDUCTION + ALTERNATING CURRENT	• Meaning of self-inductance and mutual inductance. • the L-R DC circuit (the growth and decay of current) • Energy stored in an inductor • Coefficient M and L should be understood to be defined as the constants in the equations: • $N\phi = LI$ $N\phi_2 = MI_1$ and $N\phi_1 = MI_2$	YES		04H		3.	• Familiarity with practical situations in which a single converging lens produces a magnified or diminished image, diopetre. • Prisms, optical instruments eg compound microscopes and Astronomical telescopes.	YES		02H
	20		• Theory of transformers • Knowledge, understanding but not derivation of • $\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$ for ideal transformers - Sources of power losses for practical transformers and how each is minimized.	NO		04H			• Conservation of energy for waves in free space from a point source. • Inverse square law	NO		02H
	21		• Root-mean-square values • Impedance and resonance • use of $f_0 = \frac{1}{2\pi\sqrt{LC}}$	NO		04H			• Photoelectric effect. • The photons, Plank constant	NO		02H
	22		• Power in a.c. Circuits. • Rectification of ac signals and Smoothing	NO		04H			• Einstein's photoelectric equation. • Wave-particle duality	NO		02H

	23		The atom, the nucleus of the atom,	The atom, Rutherford's alpha scattering experiment and atomic model.	NO		04H		3.3	Emission and absorption spectra. Energy levels. The electron volt.	NO		02H
	24		REVISION AND EVALUATION				04H		REVISION AND EVALUATION				02H
III	25		Radioactivity	Nuclear stability and radioactivity Properties of radiations and applications.	NO		04H	Option 2: Communication	3.3 E	Use of equation: $E = hf = E_2 - E_1$. The Schrodinger model of the hydrogen atom.	NO		02H
	26			Mass defect in nuclear processes and relation to energy Nuclear fission and fusion.	NO		04H			Representing information: Analogue method, digital method, advantages and disadvantages. Radio waves: Surface or ground wave.	NO		02H
	27	:	Option 3: Electronics	Thermionic emission and the electron gun.	YES		04H			Sky wave, space wave. Aerials: transmitting, receiving aerials. Tuning circuit, its resonance curve.	NO		02H
	28			Semiconductors and doping The p-n junction diode, biasing and applications.	NO		04H	Option 2: Communication	3.4	Transmission of information: modulation, demodulation. Amplitude modulation (AM), frequency modulation (FM), advantages and disadvantages of each over the other.	NO		02H
	29			The transistor. Logic gates and amplifiers.	NO		04H			Analogue and digital signals Analogue to Digital converters Digital to Analogue converters. Advantages of each over the other	NO		02H

			EASTER BREAK						EASTER BREAK			
30		Option 4: Medical Physics	The physics of vision and defects. Hearing and defects.	NO		04H			Main parts. Communication Channels - band width - sidebands - use of satellite for communicati on Base station and their role.	NO		02H
31			Biological Measurements for the heart. Imaging in medical diagnosis (Non-ionising methods).	NO		04H			REVISION			02H
32			Ionizing technics in imaging for medical diagnosis Use of optical fibres in medical procedures.	NO		04H			REVISION			02H
33			GENERAL REVISION			04H			GENERAL REVISION			02H
34			GENERAL REVISION			04H			GENERAL REVISION			02H
35						04H						02H
36		GENERRAL REVISION						GENERRAL REVISION				

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